

Carlos Fernando Revilla

Software Engineer | M.S. Data Science

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TECHNICAL SKILLS

Programming Languages: Python, TypeScript, JavaScript, C# (.NET), Java, C++, SQL, SAS, R

Cloud & Infrastructure: AWS, AWS cli, Terraform, Docker, CI/CD (GitHub Actions)

Data Engineering, AI/ML & Software Tools:

Apache Kafka, PostgreSQL, MongoDB, Solr, ETL/ELT Pipelines, TensorFlow, LLM Integration, Git, Jest

WORK EXPERIENCE

Philips | Full-Stack Software Engineer

Cambridge, MA | Jan 2026 - Present

- Architected a multi-service cloud platform on AWS ECS Fargate, using Terraform to provision API Gateway, ALB routing, and per-tenant state isolation, creating a scalable backbone capable of serving AI/ML model endpoints.
- Designed a fully automated CI/CD pipeline with GitHub Actions and OIDC-based AWS authentication, orchestrating Docker builds and zero-downtime deployments for multiple services.
- Built a multi-tenant C#/.NET background service with Quartz.NET and Citus/RDS to aggregate patient device data, delivering critical insights to a client-facing dashboard.
- Maintain and enhance an internal React/Vite dashboard used by clients to monitor real-time product data streams, leveraging modern AI-assisted development tools to accelerate feature delivery.

Silverchair | Full-Stack Software Engineer

Charlottesville, VA | May 2024 - February 2025

- Diagnosed and resolved critical data integrity issues in a Solr pipeline by re-ingesting missing columns, restoring content accuracy and eliminating 28% of user-reported errors.
- Identified and resolved N+1 query bottlenecks against large SQL datasets through strategic caching and batching, improving search load times by ~300ms on web and mobile apps.
- Championed code quality by building a comprehensive Jest test suite, reducing manual QA efforts by 2 hours per week for data-critical features.

WRT World Enterprises | Systems and IT Intern

Miami, FL | June 2022 – August 2022

- Built structured Excel databases to categorize and track hundreds of product details, making it easy for colleagues to quickly find specifications and answer client questions.
- Managed the product information review process, checking technical specs and descriptions against source data to maintain a high standard of quality.

EDUCATION

M.S. - Data Science | University of Virginia

Expected May 2027

- Focus: Machine learning, Data engineering, Big Data Systems, and Natural Language Processing.

B.A. - Computer Science | University of Virginia

August 2020 - May 2024

- **Minor: Data Science - Relevant Courses:** Artificial Intelligence, Data Science Systems, Database Systems

Codesmith Software Engineering Bootcamp | Full-time Immersive

May 2023 - September 2023

- **Focus:** Advanced JavaScript, TypeScript, Database Design, State Management, React + Redux Toolkit

PROJECTS

OS-Labs: KafkaNightOwl | Full-Stack Software Engineer

- Built a Message Data & Metrics Service using Node.js/Express, streamlining Kafka message transactions and persisting them via ACID-compliant MongoDB.
- Designed a scalable React/TypeScript dashboard with Flux architecture, applying container patterns for maintainable UI state management.

Acne-AI | Full-Stack Software Engineer

- Trained the DinoV2 model on a T4 GPU using Google Colab for 10 epochs, attaining train and validation loss values of 0.031 and 0.062, and achieving 99.9% model accuracy.
- Integrated a React frontend with the Python model backend using Axios, optimizing image processing and reducing diagnosis latency.

TALKS & PUBLICATIONS

GraphQL & the N+1 Problem - Speaker

July 2023

- Presented solutions to GraphQL's N+1 query issue. Talk sponsored by Codesmith, Jeeny, and Bractlet.